## Questions

**Q1.**

This simplified diagram compares spectra of light from the Sun and two galaxies.

The light from galaxy 1 and galaxy 2 both show redshift.

Explain what these redshifts predict about the position and movement of the two galaxies.

**(3)**

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**Q2.**

This question is about stars.

Use words from the box to complete the following sentences.

| | | |
|---|---|---|
| black hole | | main sequence |
| nebula | red giant | white dwarf |

(i) Stars of similar mass to our Sun were formed from a cloud of gas and dust called a ..... .

**(1)**

(ii) Our Sun is a ..... star.

**(1)**

**(Total for question = 2 marks)**

**Q3.**

Use words from the box to complete the following sentences.

| | | |
|---|---|---|
| galaxy | planet | satellite |
| solar system | star | |

(3)

(i) Saturn is a ..... .
(ii) The Moon is a ..... .
(iii) Halley's Comet orbits a ..... .

**(Total for question = 3 marks)**

**Q4.**

(i) An astronomer observes light from a distant galaxy.
As the galaxy moves away from us, the spectrum of the light is

(1)

☐ **A** blue-shifted

☐ **B** green-shifted

☐ **C** red-shifted

☐ **D** violet-shifted

(ii) The shift in the spectrum of light from the distant galaxy provides evidence for the expansion of the

(1)

☐ **A** Earth

☐ **B** Milky Way Galaxy

☐ **C** Solar System

☐ **D** Universe

**(Total for question = 2 marks)**

**Q5.**

There are many other artificial satellites in orbit around the Earth.

Figure 16 shows the time taken to complete one orbit for satellites at different heights above the Earth's surface.

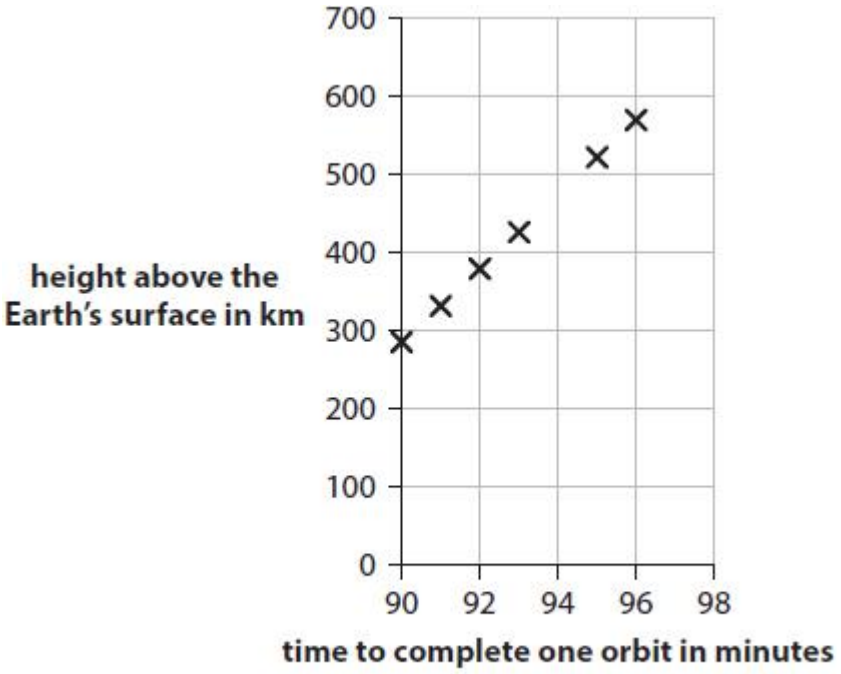


**Figure 16**

Describe the relationship shown in Figure 16.

**(2)**

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**(Total for question = 2 marks)**

**Q6.**

Describe the Solar System in terms of the Sun, the planets, and the other objects which move in the Solar System.
Your answer should include the patterns of movement of the planets and the other objects in the Solar System.
You may draw a labelled diagram if it helps your answer.

**(6)**

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**(Total for question = 6 marks)**

**Q7.**

Scientists have been searching beyond the Solar System to find planets where life could exist.

One requirement for life to exist on a planet is that there is water on the surface of the planet.

The planets where surface water could exist are in a region called the habitable zone.

Figure 12 is a diagram showing our Sun, with its five closest planets, and a more distant star with three of its planets labelled A, B and C.
The habitable zone is shaded.

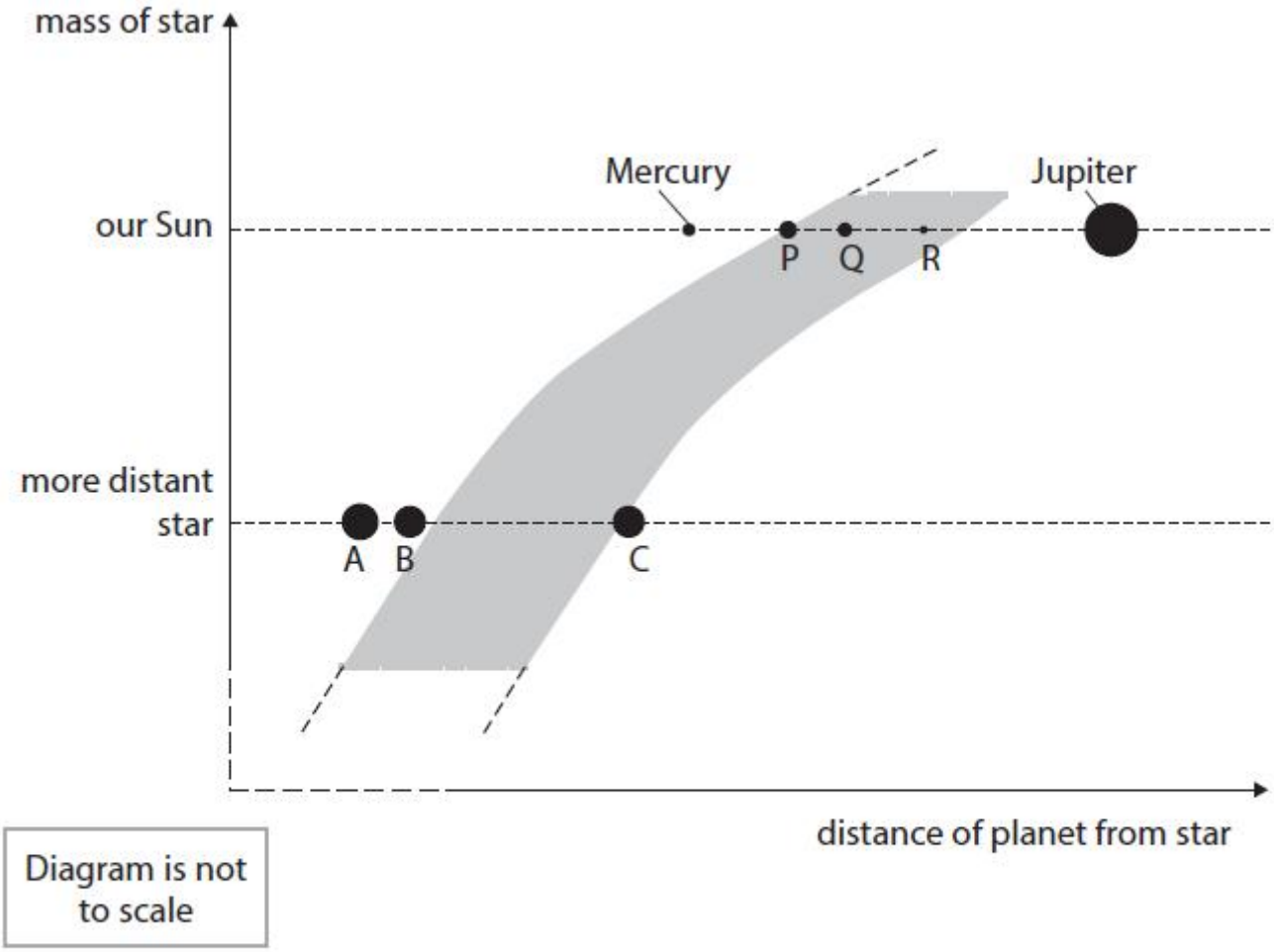


**Figure 12**

(Source adapted from: https://www.researchgate.net/publication/23993674_Darwin_-A_Mission_to_Detect_and_Search_for_Life_on_Extrasolar_Planets/figures?lo=1)

Discuss the chance of life existing on the planets shown in Figure 12.

Your answer should refer to

- the five planets shown from our Solar System, giving the names of P, Q and R.
- the three planets, A, B and C, orbiting the more distant star.

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**(Total for question = 6 marks)**

**Q8.**

Figure 1 is a table showing the distance from the Sun of the orbit of some planets.

The distances are in AU (astronomical units).

1 AU = 150 000 000 km

| planet | distance of orbit from the Sun in AU |
|---|---|
| Mercury | 0.39 |
| Earth | 1 |
| Mars | 1.5 |
| Jupiter | 5.2 |
| Neptune | 30.1 |

**Figure 1**

(i) State the distance of Earth from the Sun in kilometres.

**(1)**

distance of Earth from the Sun = ..... km

(ii) One of the planets in the table orbits the Sun between the orbits of Earth and Jupiter.
Calculate the distance from the Sun to this planet in kilometres.

**(2)**

distance from the Sun = ..... km

**(Total for question = 3 marks)**

**Q9.**

A star with a mass very much larger than the Sun

**(1)**

☐ **A** has a longer main sequence than the Sun and ends as a white dwarf

☐ **B** has a longer main sequence than the Sun and ends as a black hole

☐ **C** has a shorter main sequence than the Sun and ends as a white dwarf

☐ **D** has a shorter main sequence than the Sun and ends as a black hole

**(Total for question = 1 mark)**

**Q10.**

Figure 18 is a diagram giving some information about main sequence stars.
Luminosity is a measure of how bright something is.
An increase in luminosity means an increase in brightness.

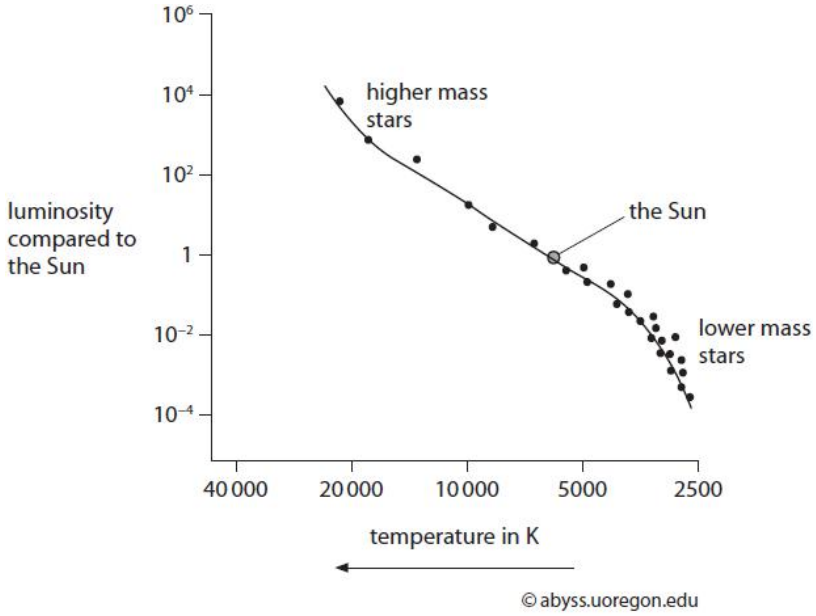


**Figure 18**

(i) Estimate the temperature of the Sun.

**(1)**

temperature of the Sun = ..... K

(ii) State how the brightness of a main sequence star changes with its temperature.

**(1)**

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(iii) State how the brightness of a main sequence star changes with its mass.

**(1)**

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**(Total for question = 3 marks)**

**Q11.**

The Big Bang theory is one theory for the origin of the Universe.

The Big Bang theory suggests:

- the Universe had a beginning
- the Universe is still expanding.

Observations of the expanding Universe have shown that the further away a galaxy is from the Earth, the faster the galaxy is moving away from the Earth.

Figure 15 shows how the velocity of galaxies is related to their distance from the Earth.

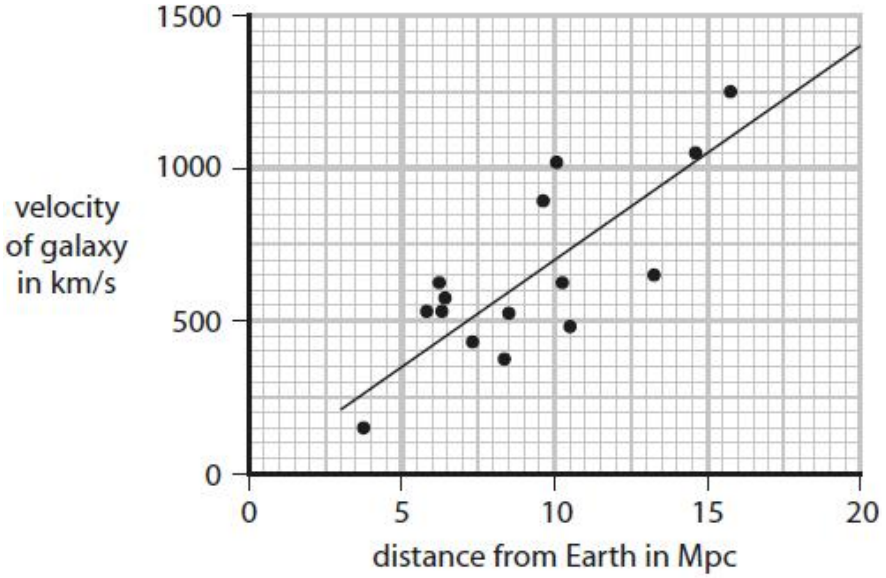


**Figure 15**

Mpc is a unit of distance used for large distances in space.

(i) Use Figure 15 to estimate the velocity of a galaxy that is 15 Mpc away from the Earth.

**(1)**

velocity = ..... km/s

(ii) Calculate the gradient of the line shown in Figure 15.

State the unit.

**(3)**

gradient = ..... unit .....

(iii) The gradient of the line in Figure 15 can be used to estimate the age of the Universe.

Explain why the gradient of the line in Figure 15 can only provide an **estimate** of the age of the Universe.

**(2)**

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**(Total for question = 6 marks)**

## Mark Scheme

Q1.

| | Answer | Acceptable answers | Mark |
|---|---|---|---|
| | An explanation linking any three of galaxies moving (1) away from Earth / Sun (1) galaxy 2 (moving away) faster (than galaxy 1) (1) galaxy 2 is (likely to be) most distant galaxy (1) | galaxies are (moving) at different speeds / away from each other / universe is expanding | **(3)** |

**Q2.**

| | Answer | Additional guidance | Mark |
|---|---|---|---|
| **(i)** | nebula (1) | allow any recognisable spelling | **(1)**<br>**AO1** |

| | Answer | Additional guidance | Mark |
|---|---|---|---|
| **(ii)** | main sequence (1) | allow any recognisable spelling | **(1)**<br>**AO1** |

**Q3.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | (i) planet (1)<br><br>(ii) satellite (1)<br><br>(iii) star (1) | in this order<br><br>accept recognisable spellings | **(3)** |

**Q4.**

| Question Number | Answer | Mark |
|---|---|---|
| (i) | **The only correct answer is C red-shifted**<br><br>A is not correct because the spectrum is not blue-shifted<br>B is not correct because the spectrum is not green-shifted<br>D is not correct because the spectrum is not violet-shifted | **(1) comp** |

| Question Number | Answer | Mark |
|---|---|---|
| (ii) | **The only correct answer is D Universe**<br><br>A is not correct because the shift does not provide evidence for the expansion of Earth.<br><br>B is not correct because the shift does not provide evidence for the expansion of the Milky Way Galaxy<br><br>C is not correct because the shift does not provide evidence for the expansion of the Solar system | **(1) comp** |

**Q5.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | a description to include<br><br>time (to complete orbit) increases as height increases (1)<br><br>relationship is linear (1) | <br><br>accept positive correlation<br>accept reverse argument<br><br>(can draw) a straight line (through the points)<br><br>accept directly proportional in this context | **2**<br>**AO3.1** |

**Q6.**

| Question number | Indicative content | Mark |
|---|---|---|
| * | Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme.<br>The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant.<br><br>**AO1**<br>• planet names: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune<br>• planet(s) orbit the Sun<br>• all planets orbit the Sun<br>• each planet in its own orbit<br>• (all) planet orbits are in the same plane<br>• planets spin (on own axis)<br>• moons orbit planets<br>• some planets have multiple moons<br>• dwarf planets orbit the Sun<br>• asteroids orbit the Sun<br>• comets orbit the Sun<br>• comet orbits not in the same plane as planets | **(6)**<br>**AO1** |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | • No rewardable material. |
| Level 1 | 1-2 | • Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1)<br>• Presents a description with some structure and coherence. (AO1) |
| Level 2 | 3-4 | • Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1)<br>• Presents a description that has a structure which is mostly clear, coherent and logical. (AO1) |
| Level 3 | 5-6 | • Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1)<br>• Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1) |

| Level | Mark | Additional Guidance | General additional guidance – the decision within levels<br><br>e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level. |
|---|---|---|---|
| | 0 | No rewardable material. | |
| Level 1 | 1–2 | Additional guidance<br><br>isolated facts about the objects or the patterns of movement | Possible candidate responses<br><br>names at least 2 planets |
| Level 2 | 3–4 | Additional guidance<br><br>Detailed description of pattern of movement for **one** set of objects.<br><br>**OR**<br><br>limited description of pattern of movement for **two** set of objects | Possible candidate responses<br><br>planets orbit the Sun<br>**AND**<br>moons orbit planets |
| Level 3 | 5–6 | Additional guidance<br><br>Detailed descriptions of pattern of movement for **two** sets of objects.<br><br>**OR**<br><br>Detailed description of pattern of movement for **one** set of objects plus limited description for **two** sets. | Possible candidate responses<br><br>all planets orbit the Sun, comets orbit the Sun, asteroids orbit the Sun |

**Q7.**

**Q8.**

| Question number | Answer | Mark |
|---|---|---|
| (i) | 150 000 000 (km) | (1) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | substitution (1)<br>1.5 x 150 000 000<br><br>evaluation (1)<br>225 000 000 (km) | sight of 1.5 or Mars named gains 1 mark<br><br>full marks will be awarded for correct numerical answer without working | (2) |

**Q9.**

| Question number | Answer | Mark |
|---|---|---|
| | B | (1) |

**Q10.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | accept any temperature between 5500 and 7500 (K) (1) | | (1) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | the higher the brightness the greater the temperature | or reverse argument<br><br>allow luminosity for brightness<br>allow heat for temperature | (1) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (iii) | the greater the mass the higher the brightness | or reverse argument<br>allow luminosity for brightness<br>allow bigger/ heavier for greater mass in this context | (1) |

**Q11.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | 1050 ± 20 (km/s) | | (1)<br>AO3<br>marked with dii |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | attempt at $\frac{\Delta y}{\Delta x}$ (1)<br><br>evaluation (1)<br><br>70 ± 5<br><br>unit (1)<br><br>km/s/Mpc | could be seen on graph<br><br>award 2 marks for correct answer without working<br><br>independent mark<br><br>km/s Mpc<br>$s^{-1}$ or per second | (3)<br>AO3 |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (iii) | an explanation linking:<br><br>points are scattered widely about the line (on graph) (1)<br><br>giving wide range of possible gradients (1) | <br><br>there are many possible best fit lines | (2)<br>**AO3** |